

EV Power - Lab 4 Project Report

Example Solution 1

Part 0: libraries

```
library(stringr)
library(dplyr)
library(ggplot2)
library(tidyr)
library(maps)
library(sf)
library(readr)
library(tidyverse)
```

Part 1: Defining Research Question I want to find which state has the cheapest energy on average (i.e. how much energy per dollar per state from 2021 to 2023, energy is measured in BTU)

Research Question: ## **Part 2: Data Preparation and Cleaning**

```
price_data_21to23 <- read.csv("av-energy-price-2021-2023.csv")
total_energy_2021 <- read.csv("total-use-2021.csv")
total_energy_2022 <- read.csv("total-use-2022.csv")
total_energy_2023 <- read.csv("total-use-2023.csv")
price_vec <- unlist(price_data_21to23)

price_vec <- str_remove(price_vec, "Total\\.energy\\.average\\.price\\.\\.dollars\\.per\\.mi.")

price_vec = str_split(price_vec, ",")
```

```

price_vec<- price_vec[-1]

price_tbl <- price_vec %>%
  map(~as_tibble_row(set_names(.x, c("State", "2021", "2022", "2023")))) %>%
  bind_rows()

price_tbl<- price_tbl|> slice(-1)

price_tbl_clean <- price_tbl |>
  mutate(
    `2021` = as.numeric(str_extract(`2021`, "\\d+\\.?\\d*")),
    `2022` = as.numeric(str_extract(`2022`, "\\d+\\.?\\d*")),
    `2023` = as.numeric(str_extract(`2023`, "\\d+\\.?\\d*"))
  )

price_2021 <- select(price_tbl_clean, State, `2021`)
price_2022 <- select(price_tbl_clean, State, `2022`)
price_2023 <- select(price_tbl_clean, State, `2023`)

energy_2023 = as_tibble(total_energy_2023)
energy_2022 = as_tibble(total_energy_2022)
energy_2021 = as_tibble(total_energy_2021)

```

Part 3: Joining / Pivoting Datasets for Analysis

```

energy_2021_long <- energy_2021 |> pivot_longer(
  cols = -Energy_Source,
  names_to = "State",
  values_to = "Energy_Use")

energy_2022_long <- energy_2022 |>
  pivot_longer(
    cols = -Energy_Source,
    names_to = "State",
    values_to = "Energy_Use")

energy_2023_long <- energy_2023 |>
  pivot_longer(

```

```

    cols = -Energy_Source,
    names_to = "State",
    values_to = "Energy_Use"
  )

avg_energy_2021 <- energy_2021_long |>
  group_by(State) |>
  summarize(avg_energy_use = mean(Energy_Use))

avg_energy_2022 <- energy_2022_long |>
  group_by(State) |>
  summarize(avg_energy_use = mean(Energy_Use))

avg_energy_2023 <- energy_2023_long |>
  group_by(State) |>
  summarize(avg_energy_use = mean(Energy_Use))

use_v_price_2021 = left_join(avg_energy_2021, price_2021)
use_v_price_2022 = left_join(avg_energy_2022, price_2022)
use_v_price_2023 = left_join(avg_energy_2023, price_2023)

head(use_v_price_2021)
head(use_v_price_2022)
head(use_v_price_2023)

```

Part 4: Mapping Visualization

```

states_map <- map_data("state")

use_v_price_2021 <- use_v_price_2021 |>
  mutate(State_full = tolower(state.name[match(State, state.abb)]))

use_v_price_2022 <- use_v_price_2022 |>
  mutate(State_full = tolower(state.name[match(State, state.abb)]))

use_v_price_2023 <- use_v_price_2023 |>
  mutate(State_full = tolower(state.name[match(State, state.abb)]))

```

```

# calculate the average consumption in thousand Btu, per dollar
btu_per_dollar_2021 <- use_v_price_2021 |>
  mutate(btu_per_dollar = avg_energy_use/`2021`)
btu_per_dollar_2022 <- use_v_price_2022 |>
  mutate(btu_per_dollar = avg_energy_use/`2022`)
btu_per_dollar_2023 <- use_v_price_2023 |>
  mutate(btu_per_dollar = avg_energy_use/`2023`)

map_2021 <- left_join(states_map, btu_per_dollar_2021, by = c("region" = "State_full"))
map_2022 <- left_join(states_map, btu_per_dollar_2022, by = c("region" = "State_full"))
map_2023 <- left_join(states_map, btu_per_dollar_2023, by = c("region" = "State_full"))
btuperdollar_map_2021 <- ggplot(map_2021, aes(x = long, y = lat, group = group, fill = btu_p
  geom_polygon(color = "white", size = 0.2) +
  coord_fixed(1.3) +
  scale_fill_gradient(low = "lightblue", high = "darkred", na.value = "grey90") +
  labs(
    title = "BTU per Dollar by State (2021)",
    fill = "BTU per $"
  ) + theme_void() + theme(
    legend.position = "bottom",
    legend.key.width = unit(2.5, "cm"),
    plot.title = element_text(hjust = 0.5, margin = margin(b = 15))
  )

btuperdollar_map_2022 <- ggplot(map_2022, aes(x = long, y = lat, group = group, fill = btu_p
  geom_polygon(color = "white", size = 0.2) +
  coord_fixed(1.3) +
  scale_fill_gradient(low = "lightblue", high = "darkred", na.value = "grey90") +
  labs(
    title = "BTU per Dollar by State (2022)",
    fill = "BTU per $"
  ) + theme_void() + theme(
    legend.position = "bottom",
    legend.key.width = unit(2.5, "cm"),
    plot.title = element_text(hjust = 0.5, margin = margin(b = 15))
  )

btuperdollar_map_2023 <- ggplot(map_2023, aes(x = long, y = lat, group = group, fill = btu_p
  geom_polygon(color = "white", size = 0.2) +
  coord_fixed(1.3) +
  scale_fill_gradient(low = "lightblue", high = "darkred", na.value = "grey90") +

```

```

labs(
  title = "BTU per Dollar by State (2023)",
  fill = "BTU per $"
) + theme_void() + theme(
  legend.position = "bottom",
  legend.key.width = unit(2.5, "cm"),
  plot.title = element_text(hjust = 0.5, margin = margin(b = 15))
)

```

```

btuperdollar_map_2021
btuperdollar_map_2022
btuperdollar_map_2023

```

#Texas by far has the greatest amount of energy use per dollar, meaning that it is the cheap